Multi Fuel Dispenser System (MFDS) with OOP





semicolon



Rules



⇒ Test the code for edge cases and write unit Test for each function

⇒ The code should be cleaned up and optimised with good naming convensions

⇒ Online outsourcing of the code is prohibited, I should see just your thought process in the code not our uncles own

Background

Digital dispensers deliver different types of petroleum products such as Petrol, Diesel. e.t.c

Each product has its own price per liter, and user may choose to buy either by specifying the number of liters or by specifying the amount of money (₦) they want to spend.

In this project, you are required to simulate a Multi-Fuel Dispenser System (MFDS) that handles these operations using OOP logics.





App Requirement

The app is made up of three entities: Fuel, Dispenser and FuelAttendant.

Fuel -> fuel_name, price_per_liter, and quantity

Dispenser -> Dictionary collection of Fuel

Fuel_Attendant -> Full name, Transactions on Dispenser



App Requirement

The Attendant should be able to do the following:

- Add a new fuel
- Get Available Fuel
- Update Fuel price
- Restock fuel





App Requirement

- Dispense fuel by Amount
- Dispense Fuel by Liters
- Show All transactions

Main Features:

1. Display available fuels and their prices per liter.
2. Allow the user to:
    - Choose fuel type.
    - Enter either:
        - Liters to buy(1 – 50 Liters), or
        - Amount (₦) to spend(must be above the price of a liter)

Main Features:

3. Calculate:
    - The total cost or total liters (depending on input).
4. Record transactions.
5. Generate a simple receipt.



THANKS!

Hope that was fun?